

1. GENERAL

1.1 TITLE: **Remote Sensing**

1.2 COURSE NUMBER: **DE.EP361.15**

1.3 CREDITS: **3-0-0 - Credit 9**

1.4 SEMESTER -OFFERED: **Odd**

1.5 Prerequisite: **Basic course on Electromagnetism**

1.6 Syllabus Committee Member: Prof. Rajendra Prasad and Prof. Prabhakar Singh

2. OBJECTIVE: Its objective is to provide basic understanding of physics and techniques for remote sensing to acquire information about land use land cover.

3. COURSE CONTENT

UNIT I: Physics of remote sensing

Overview of remote sensing, emission of electromagnetic radiation from a material body, interaction of electromagnetic radiation with atmospheric constituents and with earth's surface, Electromagnetic waves in different media, polarization and coherency of electromagnetic wave, spectral reflectance curve, electromagnetic spectrum useful for remote sensing, atmospheric windows, microwave dielectric properties of natural earth materials, remote sensing satellite orbits. (15 Lectures)

UNIT II: Sensors and platforms

Classification of sensors, active and passive sensors, scale, resolution of sensors-spectral, spatial, radiometric and temporal and mapping units, observation platforms: ground, airborne and space-borne. (6 Lectures)

UNIT III: Remote sensing antennas

Radiation mechanism of antenna, reciprocity, far field region, antenna pattern, beam dimensions, antenna directivity, antennas gain, radiation efficiency, effective area of receiving antenna, Friis transmission formula. (6 Lectures)

UNIT IV: Radar scattering models for remote sensing

Radar return from characteristic surfaces, effect of dielectric properties of targets on radar return, surface and volume scattering, influence of system and target parameters on imagery grey tones, smooth surface criteria, , scattering models for soil surfaces, angular response, roughness response and moisture response, scattering models for vegetation canopy, first order solution of radiative transfer model, first order model for multi-constituent canopy. (12 Lectures)

4. READINGS

4.1 TEXTBOOK:

1. Microwave Radar and Radiometric Remote Sensing, by F. T. Ulaby and David G. Long, The University of Michigan Press, USA.
2. Microwave Remote Sensing Active & Passive, Vol. I, II, IV by F.T. Ulaby, R.K. Moore & A.K. Fung, Addison-Wesley Publishing Company, Massachusetts. USA.

4.2 REFERENCE BOOKS:

1. Introduction to the Physics & Techniques of Remote Sensing by Charles Elachi, John Wiley & Sons, New York.
2. Remote Sensing & its Applications by L.R.A. Warayan, University Press (India) Ltd. Hyderabad

5. OUTCOME OF THE COURSE: The students would be able to understand the physical phenomenon occur in remote sensing. They may understand the sensors and antenna properties useful in remote sensing for acquiring information about distant object without coming in to contact.